

A Tasks

Task	Example
Agreement (4) agr_gender agr_sv_num_subj-relc agr_sv_num_obj-relc agr_sv_num_pp	[John/Jane] walked because [he/she] The [guard/guards] that hated the manager [is/are] The [guard/guards] that the customers hated [is/are] The [guard/guards] behind the managers [is/are]
Licensing (7) agr_refl_num_subj-relc agr_refl_num_obj-relc agr_refl_num_pp npi_any_subj-relc npi_any_obj-relc npi_ever_subj-relc npi_ever_obj-relc	The [farmer/farmers] that loved the actors embarrassed [himself/themselves] The [farmer/farmers] that the actors loved embarrassed [himself/themselves] The [farmer/farmers] behind the actors embarrassed [himself/themselves] [No/The] consultant that has helped the taxi driver has shown [any/some] [No/The] consultant that the taxi driver has helped has shown [any/some] [No/The] consultant that has helped the taxi driver has [ever/never] [No/The] consultant that the taxi driver has helped has [ever/never]
Garden path effects (6) garden_mvrr garden_mvrr_mod garden_npz_obj garden_npz_obj_mod garden_npz_v-trans garden_npz_v-trans_mod	The infant [who was/Ø] brought the sandwich from the kitchen [by/.] The infant [who was/Ø] brought the sandwich from the kitchen with a new microwave [by/.] While the students dressed [✓/Ø] the comedian [was/for] While the students dressed [✓/Ø] the comedian who told bad jokes [was/for] As the criminal [slept/shot] the woman [was/for] As the criminal [slept/shot] the woman who told bad jokes [was/for]
Gross syntactic state (4) gss_subord gss_subord_subj-relc gss_subord_obj-relc gss_subord_pp	[While the/The] lawyers lost the plans [they/.] [While the/The] lawyers who wore white lab jackets studied the book that described several advances in cancer therapy [✓/.] [While the/The] lawyers who the spy had contacted repeatedly studied the book that colleagues had written on cancer therapy [✓/.] [While the/The] lawyers in a long white lab jacket studied the book about several recent advances in cancer therapy [✓/.]
Long-distance dependencies (8) cleft cleft_mod filler_gap_embed_3 filler_gap_embed_4 filler_gap_hierarchy filler_gap_obj filler_gap_pp filler_gap_subj	What the young man [did/ate] was [make/for] What the young man [did/ate] after the ingredients had been bought from the store was [make/for] I know [that/what] the mother said the friend remarked the park attendant reported your friend sent [him/.] I know [that/what] the mother said the friend remarked the park attendant reported the cop thinks your friend sent [him/.] The fact that the brother said [that/who] the friend trusted [the/was] I know [that/what] the uncle grabbed [him/.] I know [that/what] the uncle grabbed food in front of [him/.] I know [that/who] the uncle grabbed food in front of [him/.]

B Training and evaluation details

We load models using the HuggingFace transformers (Wolf et al., 2020) library. Up to size 410m we load weights in float32 precision, 1b in bfloat16 precision, and larger models in float16 precision. Our training set starts with 200 examples sampled according to the scheme in §3.2. We then double the size of the set (400) by swapping the base and source inputs/labels and adding these to the training set; including both directions of the intervention makes the comparison fairer between DAS and the other non-paired methods, and also ensures a perfect balance between labels.

The evaluation set consists of 50 examples sampled the same way (effectively 100), except we resample in case we encounter a sentence already present in the training set. Thus, there is no overlap with the training set. We evaluate all metrics (odds-ratio and probe classification accuracy) on this set.

We train DAS for one epoch with a batch size of 4, resulting in 100 backpropagation steps. We use the Adam optimiser (Kingma and Ba, 2015) and a linear learning rate schedule, with the first 10% of training being a warmup from 0 to the learning rate, followed by the learning rate linearly decaying to 0 for the rest of training. The scheduling and optimiser is identical to Wu et al. (2023). We use a learning

rate of $5 \cdot 10^{-3}$, which is higher than previous work (usually 10^{-3}) due to the small training set size; see appendix C for hyperparameter tuning experiments which justify this choice.

To run our experiments, we used a cluster of NVIDIA A100 (40 GB) and NVIDIA RTX 6000 Ada Generation GPUs. The total runtime for the benchmarking experiments in §5 was ~ 400 hours, and for the case studies in §6 it was ~ 25 hours.

C Hyperparameter tuning

To ensure fair comparison, we tuned hyperparameters for DAS, probes, and PCA on a dev set, sampled the same way as the eval set (non-overlapping with train set) but with a different random seed. We train on all tasks in **CausalGym** and report the average odds-ratio following the same evaluation setup as in §5.1. We studied only the three smallest models (pythia-14m, 31m, 70m) due to the large number of experiments needed. Specifically, we tune the learning rate for DAS, the type of regularisation and whether or not to include a bias term in the logit for probes,³ and averaging of the first c components for PCA. We report the overall log odds-ratio for various hyperparameter settings in Table 2. These experiments were run on a NVIDIA RTX 6000 Ada Generation. The total runtime was ~ 25 hours.

For probing (Table 2a), we found that including a bias term and using only L_2 regularisation with the saga solver delivers the best performance. However, the setting of the weight coefficient λ on the regularisation term in the loss depends on the model. The main architectural difference between these three models is the hidden dimension size, so we suspect that the optimal choice for λ depends on that. Roughly extrapolating the observed trend, in our main experiments we check $\lambda = \{10^4, 10^5\}$ for pythia-160m and 410m, $\lambda = \{10^5, 10^6\}$ for pythia-1b, 1.4b, and 2.8b, and $\lambda = \{10^6, 10^7\}$ for pythia-6.9b. As for why L_2 regularisation increases causal efficacy, we note that Hewitt and Liang (2019) found that it also increases probe selectivity—we leave this as an open question for future work.

For PCA (Table 2b), we found that averaging the first c components did not improve performance over just using the first component; thus, we used just the first PCA component in our main-text experiments.

For DAS (Table 2c), we found that using the learning rate suggested by Wu et al. (2023), 10^{-3} , understated performance and a higher learning rate did not result in any apparent training instability. However, our experimental setup is quite different (smaller training set, no learned boundary, greater variety of model scales). We did not find any consistent differences or trends with model scale between learning rates of $5 \cdot 10^{-3}$ and 10^{-2} , so we used the former for all experiments.

D Data and licensing

We use the original test suites from SyntaxGym which were described in Hu et al. (2020). These were released under the MIT License, and our data release will also use the MIT license for compatibility.

³Cf. Tigges et al. (2023), who did not include a bias term in their causal evaluation of probing.

Model	Probe	λ				
		10^0	10^1	10^2	10^3	10^4
pythia-14m ($d = 128$)	No reg., no int.	0.80				
	No reg., int.	0.85				
	L_1 , no int.	0.38	0.21		0.00	
	L_1 , int.	0.41	0.22		0.00	
	L_2 , no int.	1.07	1.15		1.08	
	L_2 , int.	1.09	1.18	1.15	1.07	1.05
	$L_1 + L_2$, no int.	0.93	0.55		0.08	
	$L_1 + L_2$, int.	0.89	0.57		0.08	
pythia-31m ($d = 256$)	No reg., no int.	1.75				
	No reg., int.	1.77				
	L_1 , no int.	0.83	0.39		0.12	
	L_1 , int.	0.83	0.40		0.11	
	L_2 , no int.	1.98	2.14		2.18	
	L_2 , int.	2.03	2.22	2.26	2.11	1.90
	$L_1 + L_2$, no int.	1.45	0.99		0.14	
	$L_1 + L_2$, int.	1.42	0.93		0.14	
pythia-70m ($d = 512$)	No reg., no int.	1.72				
	No reg., int.	1.72				
	L_1 , no int.	0.74	0.32		0.31	
	L_1 , int.	0.75	0.33		0.17	
	L_2 , no int.	1.85	2.05		2.43	
	L_2 , int.	1.87	2.08	2.38	2.70	2.57
	$L_1 + L_2$, no int.	1.11	0.67		0.32	
	$L_1 + L_2$, int.	1.12	0.71		0.18	

(a) Overall odds-ratio across various hyperparameter settings for probes. ‘Int.’ means whether the probe logit has a bias term.

Model	c (# components)				
	1	2	3	4	5
pythia-14m	0.48	0.44	0.34	0.29	0.28
pythia-31m	0.86	0.82	0.59	0.49	0.43
pythia-70m	1.18	0.91	0.78	0.75	0.64

(b) Overall odds-ratio across variants of PCA, averaging the first c components.

Model	LR	Step				
		0	25	50	75	99
pythia-14m	10^{-3}	0.06	0.37	1.01	1.48	1.63
	$5 \cdot 10^{-3}$	0.04	2.53	3.58	3.82	3.91
	10^{-2}	0.04	3.17	3.72	3.95	4.02
pythia-31m	10^{-3}	0.04	1.09	2.83	3.64	3.83
	$5 \cdot 10^{-3}$	0.04	5.19	5.78	6.00	6.04
	10^{-2}	0.03	5.05	5.44	5.77	5.87
pythia-70m	10^{-3}	0.02	2.25	4.69	5.42	5.57
	$5 \cdot 10^{-3}$	0.02	7.21	7.48	7.54	7.55
	10^{-2}	0.03	6.92	7.37	7.66	7.75

(c) Overall odds-ratio across various learning rates for DAS.

Table 2: Hyperparameter search results.

E Detailed odds-ratio results

In these comprehensive results, we include an additional method: vanilla interchange intervention. Instead of as in eq. (11), vanilla intervention defines f^* as

$$f_{\text{vanilla}}^*(\mathbf{b}, \mathbf{s}) = f(\mathbf{s}) \quad (24)$$

i.e. it entirely replaces the activation with that of the source input. This is equivalent to n -dimensional DII where $f(\mathbf{s}) \in \mathbb{R}^n$, and is a significantly more expressive intervention than any methods we tested.

E.1 Per-layer

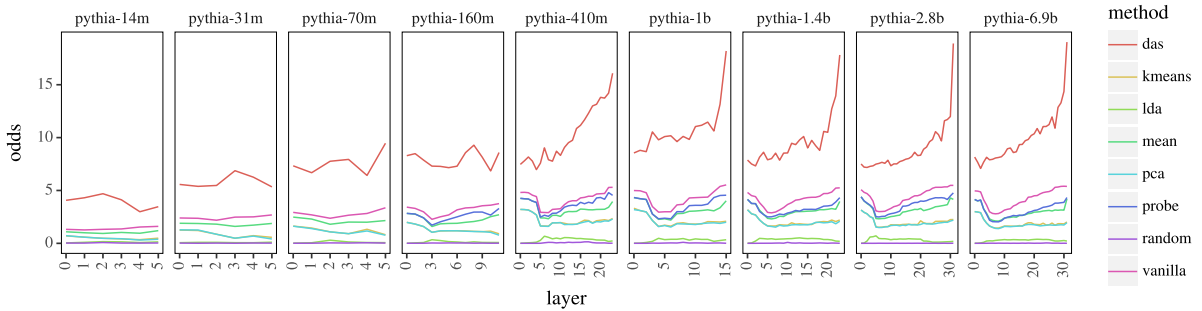


Figure 7: Average odds-ratio per layer and model across all tasks in CausalGym.